Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/30/2025
PI Name:	Rice, Kelly Ph.D.		
Project Title:	Alteration of Bacillus subtilis DNA Architecture in Space: Global Effects on DNA Supercoiling, Methylation, and the Transcriptome		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611-0700	**Congressional District:**	3
Comments:			
Project Type:		**Solicitation / Funding Source:**	03-OBPR-02
Start Date:	12/03/2020	**End Date:**	12/02/2025
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	O'Rourke, Aubrie	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: Start date changed to 12/03/2020 per NASA-KSC (Ed., 6/16/25). NOTE: End date changed to 12/02/2025 per NSSC information (Ed., 12/19/24). NOTE: End date changed to 12/02/2024 per NSSC information (Ed., 1/15/24).		
Key Personnel Changes/Previous PI:	Project is a continuation of "Alteration of Bacillus subtilis DNA Architecture in Space: Global Effects on DNA Supercoiling, Methylation, and the Transcriptome". Previous PI: Dr. Wayne Nicholson. NOTE: Dr. Kelly Rice became Principal Investigator for this project August 21, 2023 (Ed., 11/20/23).		
COI Name (Institution):	Fajardo-Cavazos, Patricia Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC21K0319		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	1) Description of the research product(s): Studying the response of microbial cells to the spaceflight environment is a high-priority issue in Space Biology, due to the importance of microbes in maintaining the health of astronauts and their habitats. Previous International Space Station (ISS) flight experiments (missions Biological Research in Canisters (BRIC)-18, -21, and -23)) using the Gram-positive model bacterium Bacillus subtilis (B. subtilis) have uncovered global changes in the transcriptome, altered mutagenic spectrum of DNA, and significant differences in cell susceptibility to fluoroquinolone antibiotics in spaceflight vs. ground controls. Fluoroquinolones target the enzymes DNA gyrase and topoisomerase IV, which are critical for maintaining DNA supercoiling as well as nucleoid compaction and conformation. Changes in DNA architecture affect many downstream genetic events such as: epigenetic DNA modifications (i.e., the methylome); the location of mutagenic hotspots in DNA; and global gene transcription patterns (i.e., the transcriptome). In turn, such alterations in global gene expression may lead to phenotypic changes observed in spaceflight. Analyses of the data from our prior missions has led us to formulate the following working hypothesis that can be tested experimentally using the BRIC-B. subtilis experimental system: Hypothesis: Exposure to spaceflight microgravity leads to changes in nucleoid architecture, DNA supercoiling, methylation, global transcription patterns, and resulting phenotypes. 2) Type of investigation: ISS with ground controls. 3) Specific aims: Using the B. subtilis model system, compare global: (i) 3-dimensional nucleoid architecture, (ii) DNA supercoiling, (iii) DNA methylation, and (iv) transcriptomes in spaceflight vs. matched ground controls. 4) Experimental plan: Experiments will utilize BRIC-Petri Dish Fixation Unit (PDFU) hardware modified with newly designed inserts mitigating the confounding effect of microgravity on liquid geometry. Four global parameters will be measured: (i) 3-dimensional nucleoid architecture using chromosome conformation capture (Hi-C); (ii) DNA supercoiling using 2-dimensional chloroquine-agarose gel electrophoresis; (iii) DNA methylation using Methyl-seq; and (iv) transcriptomes using RNA-seq. Results will probe at the molecular level the effect of spaceflight microgravity on four fundamental biological processes and their interactions. 5) NASA Relevance: This proposal directly addresses the Microbiology and the Cell and Molecular Biology Elements of the ROSBio-2018 Omnibus announcement and the NASA Strategic Plan Goals, especially Objective 1.2. The proposal directly addresses NASA's Space Biology Science Plan for 2016-2025, most particularly the overarching Microbiology Element Goal "to determine the effect of spaceflight on microbial life, processes, and community dynamics," specifically: 1) MB-1: What underlying genetic, molecular and biochemical processes are influenced by the space environment? (P2); 2) MB-2: What systems biology mechanisms and pathways for microbial life are responsive to spaceflight conditions? (P1, P2); 3) MB-4: What are the mechanisms that effect changes such as the altered virulence or altered drug resistance observed in some organisms during spaceflight? (P1, P2, P3); and 4) MB-5: How does the spaceflight environment influence microbial reproduction, growth, and physiology? (P1, P2).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal of the project is to understand at a fundamental (molecular) level how bacteria sense and respond to the spaceflight environment. At present it is known that the phenotypic behavior of microbes is altered when confronted with spaceflight microgravity, but to date the underlying mechanism(s) involved have remained obscure. The project is to measure four fundamental molecular processes in cells of the model bacterium Bacillus subtilis: three-dimensional nucleoid architecture, DNA supercoiling, DNA methylation, and transcriptomes. These processes will be measured and compared in spaceflight (FL) vs. matched ground control (GC) samples. Deeper understanding of the molecular responses of bacteria to prolonged spaceflight has important implications for understanding disease processes and performance of microbes in bioreactors both in space and on Earth.
Task Progress:	Our spaceflight experiment to the International Space Station (ISS), designated Biological Research in Canisters-26 (BRIC-26), launched on SpaceX CRS-26 on November 26, 2022. The BRIC-26 payload was successfully actuated, incubated, and stored in the onboard -80°C Minus Eighty (Degrees Celsius) Laboratory Freezer for ISS (MELFI) freezer for return to Earth. This payload was returned to Earth, retrieved from stowage on or around 01/11/23. Samples were de-integrated at NASA Kennedy Space Center (KSC) on 03/16/2023 and transferred to the Nicholson lab's -80 C freezer at the Space Life Sciences Lab (SLSL). HOBO® data from the BRIC-26 flight data was downloaded and used to program the ISS incubation chamber at KSC for the asynchronous ground control (GC) experiment which was performed in January 2024 (integration on 01/09/2024, actuation on 01/26/2024, and de-integration on 02/07/2024). Most flight (FLT) and GC samples were grown in "gizmo" petri dishes, designed to mitigate the influence of microgravity on liquid geometry in BRIC canisters, as it was noted that in previous BRIC flight experiments, the liquid cultures displayed a toroidal ("donut") shape in regular petri dishes compared with the ground control samples (a flat disc shape). BRIC-26 results were consistent with previous BRIC flight experiments, whereby FLT regular petri dish liquid cultures displayed a donut-like shape, and GC cultures maintained a flat disc shape. Although gizmo FLT liquid cultures did maintain a flat disc-like shape relative to the donut shape observed in petri FLT liquid cultures, the gizmo plate FLT liquid cultures only covered ~1/3 to 2/3 of the total surface area of the gizmo plate. As expected, the gizmo GC liquid cultures maintained a disc shape that covered the entire plate. RNA was isolated from n=4 each of gizmo FLT and GC samples, and n=3 each of petri FLT and GC2 samples, with excellent RNA yields and purity. Ribodepletion, library preparation, sample barcoding, and Illumina sequencing were performed by the University of Florida (UF)'s Interdisciplinary Center for Biotechnology Research (ICBR). Hierarchical clustering analysis revealed overall strong clustering among replicates in each growth condition. There was a 50% overlap in transcriptome changes between BRIC-26 Gizmo FLT/GC and Petri FLT/GC comparisons. Bioinformatics analysis is ongoing to identify gene enrichments specific to gizmo and petri FLT cultures, as well as identifying the overlapping transcriptome between BRIC-26 and previous B. subtilis flight experiments (i.e., BRIC-21). Genomic DNA was also isolated from n=3 each of gizmo FLT and GC samples, with excellent yields and purity to proceed with PacBio Revio sequencing (conducted at UF-ICBR). Methylations (5mC, 6mA, 4mC) were identified for each sample (analysis provided by the UF-ICBR core). Analysis of this data is ongoing and will include differential methylation analysis (DMA) as well as correlative analysis with the RNA-seq data described above. This will determine if transcriptome changes in FLT correlate with changes in DNA methylation at the relevant promoter sites.

Bibliography Type:	Description: (Last Updated: 02/19/2026)